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CYS233 Project

School network

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Introduction

Background

Organizational networks are crucial for communication within a company. They help manage the flow of information, ensuring that data gets from one person to another efficiently. A good network connects multiple devices, whether through wires or wirelessly.

Networks are important in both work and personal life. They allow us to share files easily, improve efficiency, and provide flexibility in how we communicate. As businesses grow in a digital world, strong organizational networks are essential for teamwork, innovation, and staying competitive.

Objective of the Network

The primary web server, classroom's locations, and the network room are all connected by a built network topology. That was designed by using CiscoPacket Tracer 8.2. The goal of this design is to provide a clear link between these parts, enabling efficient communication and data exchange. This will makesure that whenever a teacher is trying to access the network he will connect safely to the school's internet. This network will make use of different networking components like routers, switches, phones, and printers. These elements will collaborate seamlessly to guarantee smooth communication.

Topology Requirements

The topology includes the following elements:

PC	7
Laptop	6
Printer-PT	3
Cisco Switch	5
Cisco Router	3
IP Phone	3
Server	1
DSL Modem	1
Cloud-PT	1
Access point	2

Topology

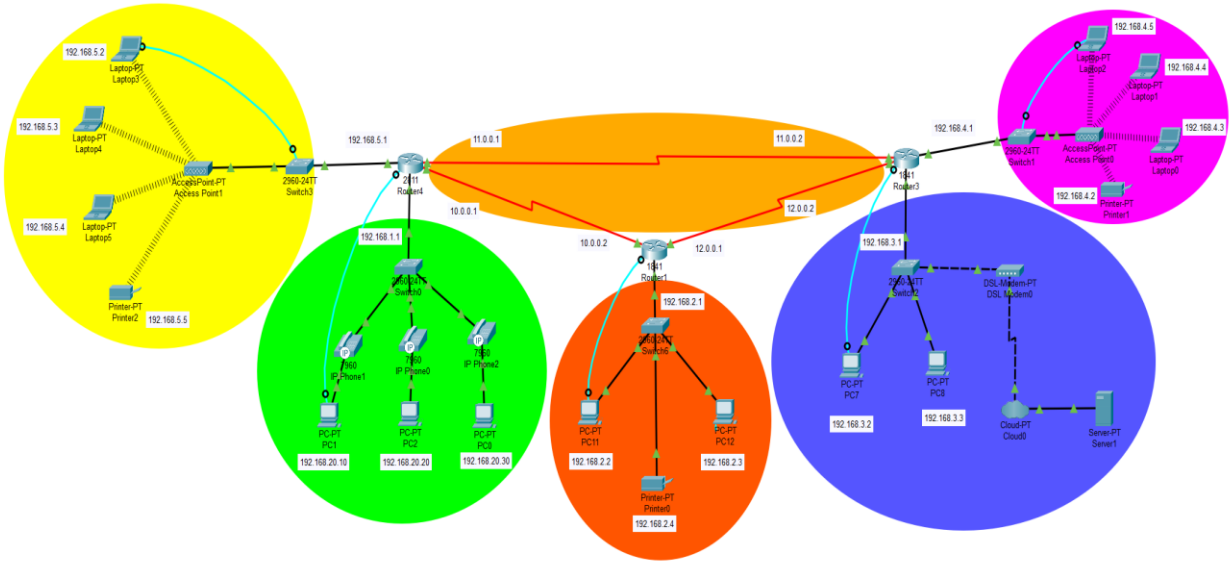

Figure 1: Topology image

Addressing/Ports Table

Device	Interface	IPv4 Address	Subnet Mask	Default GateWay
(Router1)	FastEthernet0/0	192.168.2.1	255.255.255.0	N/A
	Serial0/1/0	10.0.0.2	255.0.0.0	N/A
	Serial0/1/1	12.0.0.1	255.0.0.0	N/A
(Router3)	FastEthernet0/0	192.168.3.1	255.255.255.0	N/A
	FastEthernet0/1	192.168.4.1	255.255.255.0	N/A
	Serial0/1/0	11.0.0.2	255.0.0.0	N/A
	Serial0/1/1	12.0.0.2	255.0.0.0	N/A
(Router4)	FastEthernet0/0	192.168.1.1	255.255.255.0	N/A
	FastEthernet0/1	192.168.5.1	255.255.255.0	N/A
	Serial0/0/0	10.0.0.1	255.0.0.0	
	Serial0/1/1	11.0.0.1	255.0.0.0	N/A
IP Phone0 Line Number: 12345	Vlan10	192.168.10.3	255.255.255.0	192.168.10.1
IP Phone1 Line Number: 54321	Vlan10	192.168.10.4	255.255.255.0	192.168.10.1
IP Phone2 Line Number: 67890	Vlan10	192.168.10.2	255.255.255.0	192.168.10.1
Printer0	FastEthernet0	192.168.2.4	255.255.255.0	192.168.2.1
Printer1	Wireless0	192.168.4.2	255.255.255.0	192.168.4.1

Printer2	Wireless0	192.168.5.5	255.255.255.0	192.168.5.1
PC0	FastEthernet0	192.168.20.30	255.255.255.0	192.168.20.1
PC1	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.1
PC2	FastEthernet0	192.168.20.20	255.255.255.0	192.168.20.1
PC7	FastEthernet0	192.168.3.2	255.255.255.0	192.168.3.1
PC8	FastEthernet0	192.168.3.3	255.255.255.0	192.168.3.1
Pc11	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
Pc12	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1
Server1	FastEthernet0	192.168.3.4	255.255.255.0	192.168.3.1
Cloud-PT	Moderm4	N/A	N/A	N/A
	Ethernet6	N/A	N/A	N/A

Router's Configuration

Methods

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname group1
group1(config)#enable secret class
group1(config)#line console 0
group1(config-line)#password group1
group1(config-line)#login
group1(config-line)#line vty 0 4
group1(config-line)#password group1
group1(config-line)#login
group1(config-line)#exit
group1(config)#service password-encryption
group1(config)#banner MOTD # 224  ,  group1#
group1(config)#end
group1#
%SYS-5-CONFIG I: Configured from console by console
```

Applied Configuration

Enable secret: **class**

Line console and vty password: **group1**

(Router1):

Network Address
192.168.3.0/24 via 12.0.0.2
192.168.20.0/24 via 10.0.0.1
192.168.10.0/24 via 10.0.0.1
192.168.4.0/24 via 12.0.0.2
192.168.5.0/24 via 10.0.0.1

Figure 2: Router1 Network Addresses

```
Device Name: Router1
Device Model: 1841
Hostname: group1

Port              Link   VLAN   IP Address         IPv6 Address                              MAC Address
FastEthernet0/0   Up     --     192.168.2.1/24     <not set>                                 000C.8591.5301
FastEthernet0/1   Down   --     <not set>          <not set>                                 000C.8591.5302
Serial0/1/0       Up     --     10.0.0.2/8         <not set>                                 <not set>
Serial0/1/1       Up     --     12.0.0.1/8         <not set>                                 <not set>
Vlan1             Down   1      <not set>          <not set>                                 0010.1144.6A89

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1
```

Figure 3: Router1 Interfaces

(Router3):

Network Address
192.168.2.0/24 via 12.0.0.1
192.168.20.0/24 via 11.0.0.1
192.168.10.0/24 via 11.0.0.1
192.168.5.0/24 via 11.0.0.1

Figure 4: Router3 Network Address

```
Device Name: Router3
Device Model: 1841
Hostname: group1

Port              Link   VLAN   IP Address         IPv6 Address                          MAC Address
FastEthernet0/0   Up     --     192.168.3.1/24     <not set>                             00D0.58EE.EC01
FastEthernet0/1   Up     --     192.168.4.1/24     <not set>                             00D0.58EE.EC02
Serial0/1/0       Up     --     11.0.0.2/8         <not set>                             <not set>
Serial0/1/1       Up     --     12.0.0.2/8         <not set>                             <not set>
Vlan1             Down   1      <not set>          <not set>                             0000.0CE1.3610

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router3
```

Figure 5: Router3 Interfaces

(Router4):

Network Address
192.168.2.0/24 via 10.0.0.2
192.168.3.0/24 via 11.0.0.2
192.168.4.0/24 via 11.0.0.2

Figure 6: Router4 Network Address

```
Device Name: Router4
Custom Device Model: 2811 IOS15
Hostname: group1

Port                Link   VLAN   IP Address         IPv6 Address                          MAC Address
FastEthernet0/0     Up     --     192.168.1.1/24     <not set>                             0001.63C0.6701
FastEthernet0/0.10  Up     --     192.168.10.1/24    <not set>                             0001.63C0.6701
FastEthernet0/0.20  Up     --     192.168.20.1/24    <not set>                             0001.63C0.6701
FastEthernet0/1     Up     --     192.168.5.1/24     <not set>                             0001.63C0.6702
Serial0/0/0         Up     --     10.0.0.1/8         <not set>                             <not set>
Serial0/0/1         Up     --     11.0.0.1/8         <not set>                             <not set>
Vlan1               Down   1      <not set>          <not set>                             0001.96E6.2AD5

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router4
```

Figure 7: Router4 Interfaces

Switch Configuration

Methods

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname group1
group1(config)#enable secret class
group1(config)#line console 0
group1(config-line)#logging synchronous
group1(config-line)#password group1
group1(config-line)#login
group1(config-line)#line vty 0 14
group1(config-line)#password group1
group1(config-line)#login
group1(config-line)#exit
group1(config)#service pass
group1(config)#service password-encryption
group1(config)#banner MOTD # 224 , group1#
group1(config)#no ip domain-lookup
group1(config)#end
group1#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 8: Switch Configurations

Applied Configuration

Enable secret: **class**

Line console and vty password: **group1**

VoIP Configuration

IP Phone 0:

```
Device Name: IP Phone1
Device Model: 7960

Port     Link   IP Address         MAC Address
Vlan1    Up     192.168.2.3/24     0002.161A.00A1
Switch   Up     <not set>          0005.5E8C.1B01
PC       Up     <not set>          0005.5E8C.1B02

Gateway: 192.168.2.1
Line Number: 11111

Physical Location: Intercity > Home City > Corporate Office > IP Phone1
```

Figure 9: IP Phone0 interfaces

IP Phone 1:

```
Device Name: IP Phone1
Device Model: 7960

Port     Link   IP Address         MAC Address
Vlan1    Down   <not set>          00E0.F7BA.C267
Switch   Up     <not set>          00D0.D37D.5101
PC       Up     <not set>          00D0.D37D.5102
Vlan10   Up     192.168.10.4/24    00E0.F7BA.C267

Gateway: 192.168.10.1
Line Number: 54321

Physical Location: Intercity > Home City > Corporate Office > IP Phone1
```

Figure 10: IP Phone1 interfaces

IP Phone 2:

```
Device Name: IP Phone2
Device Model: 7960

Port     Link   IP Address         MAC Address
Vlan1    Down   <not set>          0001.9730.E7D4
Switch   Up     <not set>          0010.113B.4501
PC       Up     <not set>          0010.113B.4502
Vlan10   Up     192.168.10.2/24    0001.9730.E7D4

Gateway: 192.168.10.1
Line Number: 67890

Physical Location: Intercity > Home City > Corporate Office > IP Phone2
```

Figure 11: IP Phone2 interfaces

PC Configuration

PC0:

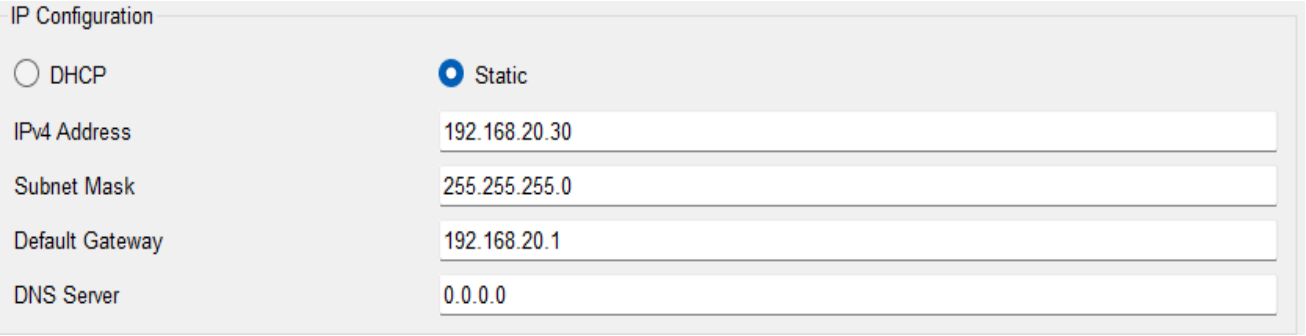

Figure 12: PC0 IP Configurations

PC1:

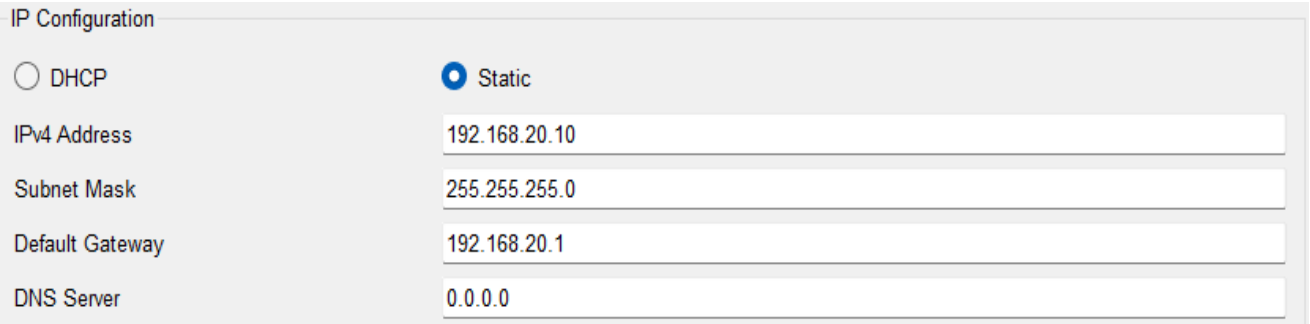

Figure 13: PC1 IP Configurations

PC2:

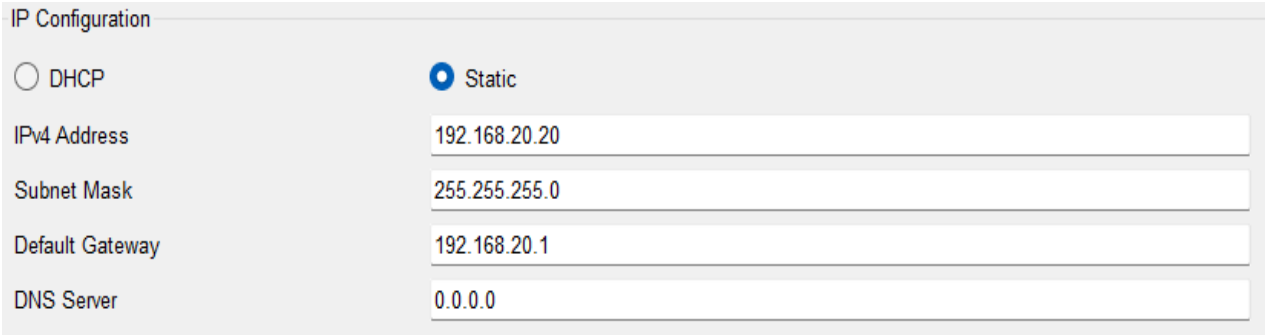

Figure 14: PC2 IP Configurations

PC7:

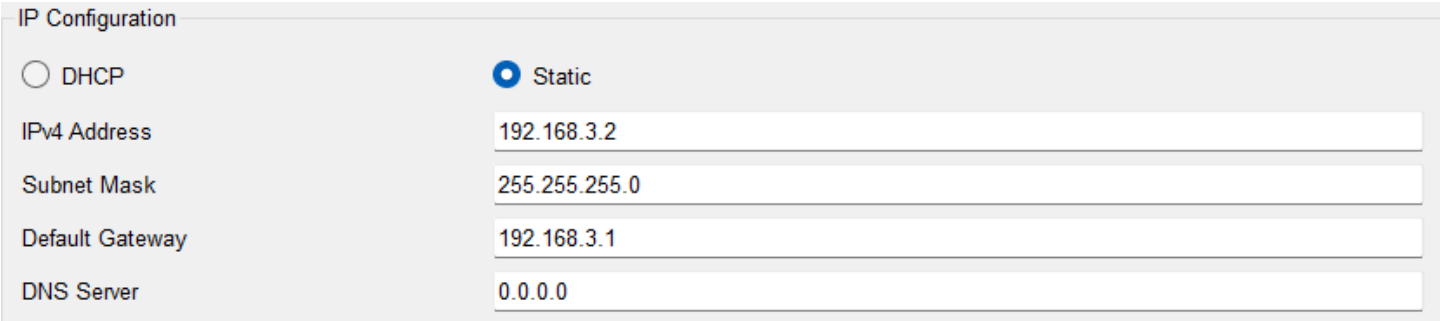

Figure 15: PC7 IP Configurations

PC8:

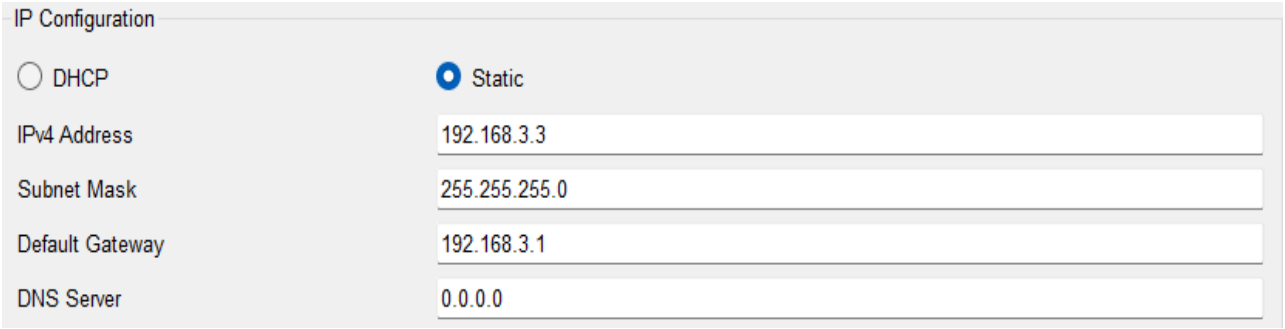

Figure 16: PC8 IP Configurations

Pc11:

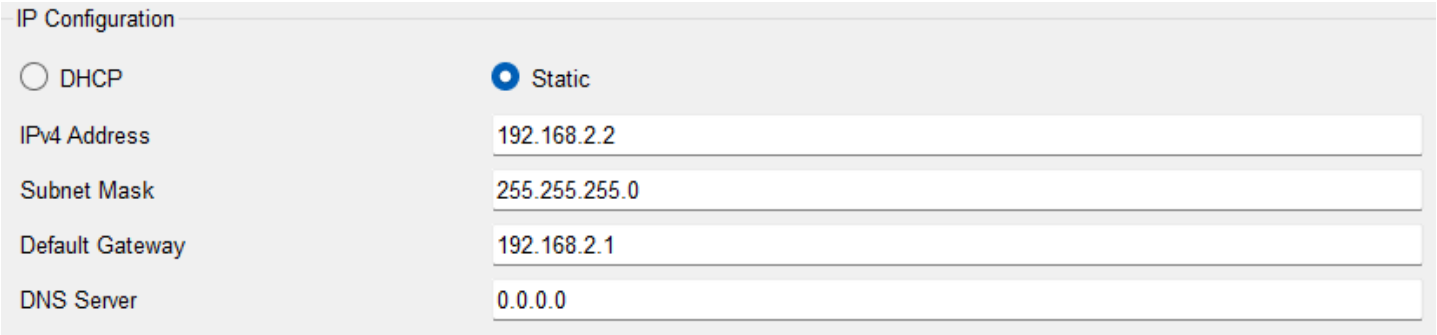

figure 17: pc11 IP configurations

Pc12:

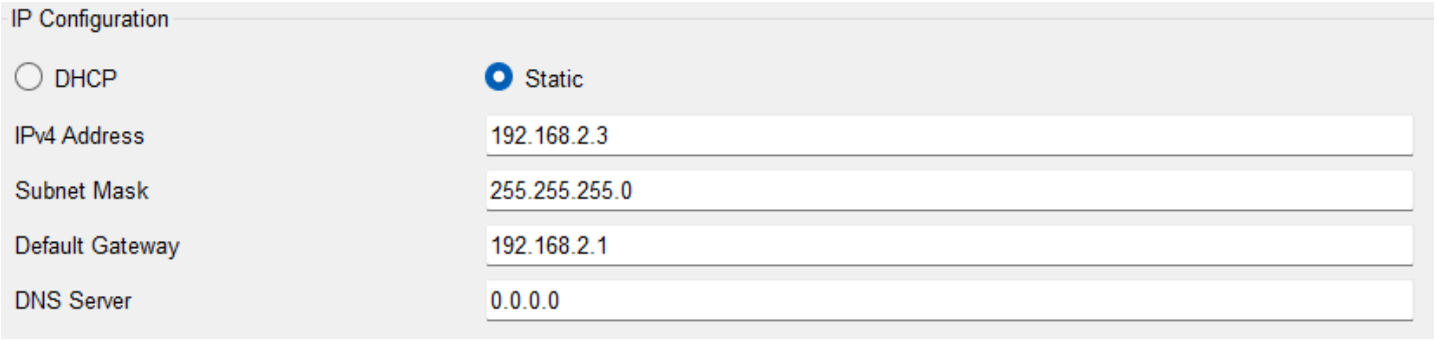

figure 18: pc12 IP configurations

Printer Configuration

Printer0:

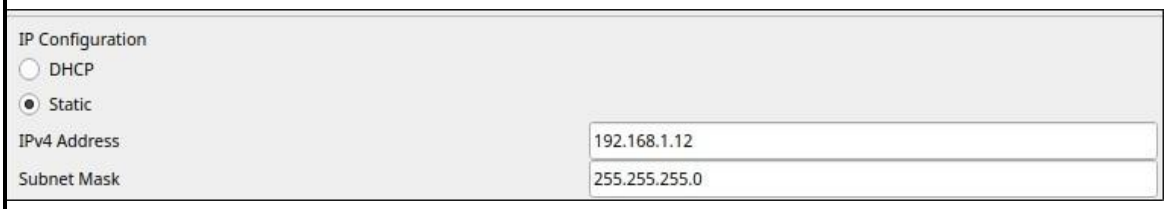

Figure 19: Printer0 IP Configuration

Figure 20: Printer0 GateWay Configuration

Printer1:

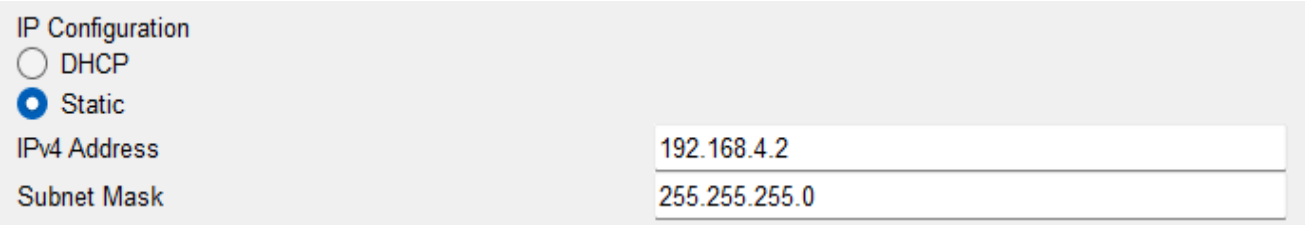

Figure 21: Printer1 IP Configuration

Figure 22: Printer1 GateWay configuration

Servers Configuration

Server1:

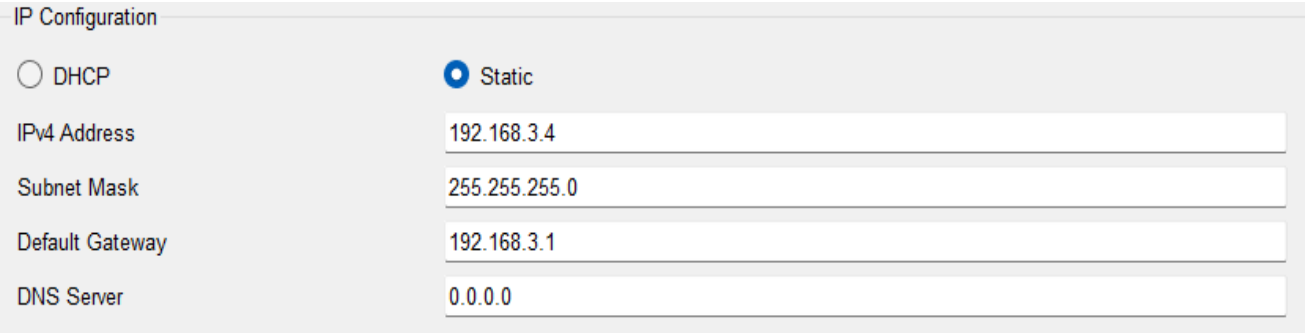

Figure 23: Server1 IP Configuration

DSL Modem Configuration

Cloud0:

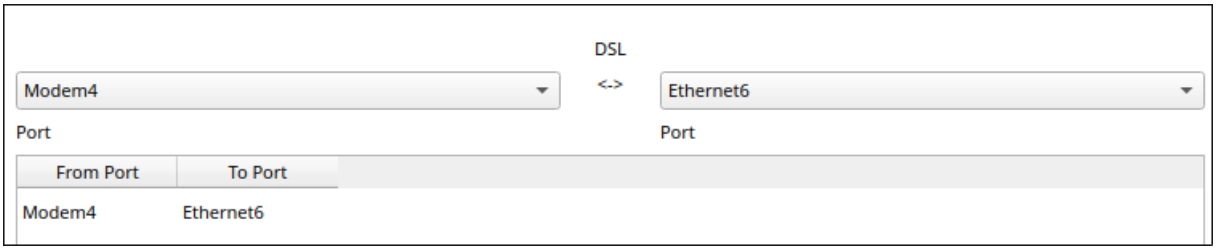

Figure 24: Cloud0 Ports Configuration

Verification

IP Phones connected:

Figure 25: VoIP test

Every network is connected:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:47FF:FEDD:9611
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.20.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time=11ms TTL=126
Reply from 192.168.3.2: bytes=32 time=12ms TTL=126
Reply from 192.168.3.2: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 12ms, Average = 10ms

C:\>
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:47FF:FEDD:9611
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.20.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=14ms TTL=126
Reply from 192.168.2.2: bytes=32 time=13ms TTL=126
Reply from 192.168.2.2: bytes=32 time=5ms TTL=126
Reply from 192.168.2.2: bytes=32 time=13ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 14ms, Average = 11ms
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:47FF:FEDD:9611
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.20.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.4.3

Pinging 192.168.4.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.3: bytes=32 time=16ms TTL=126
Reply from 192.168.4.3: bytes=32 time=23ms TTL=126
Reply from 192.168.4.3: bytes=32 time=32ms TTL=126

Ping statistics for 192.168.4.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 32ms, Average = 23ms
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:47FF:FEDD:9611
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.20.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time=19ms TTL=127
Reply from 192.168.5.2: bytes=32 time=15ms TTL=127
Reply from 192.168.5.2: bytes=32 time=16ms TTL=127
Reply from 192.168.5.2: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 15ms, Maximum = 19ms, Average = 16ms
```

Figure 26: pinging (working)

Conclusion

In conclusion, organizational networks are essential for promoting effective communication within a company. They enable smooth information transfer between people and devices, regardless of whether the connections are wired or wireless. These networks are crucial not only in professional environments but also improve personal communication and data sharing capabilities.

In the end, a well-designed organizational network is essential for enhancing teamwork, driving innovation, and maintaining a competitive advantage in today's digital organizations.